

Gröbner bases

(Taken from Dummit and Foote, Abstract Algebra, §9.6)

For the entirety of this document, we will be working in the polynomial ring $R = F[x_1, \dots, x_n]$ where F is some field.

Definition 1. (Monomial ordering)

A monomial ordering is a well-ordering $\geq$ on the set of (monic) monomials such that $m_1 \geq m_2$ implies $mm_1 \geq mm_2$. Note that the well-ordering implies that $m \geq 1$ for all m .

In fact, any total ordering on monomials is a well-ordering if and only if it satisfies $m \geq 1$ and $mm_1 \geq mm_2$.

Equivalently, a monomial ordering is specified by a well-ordering on $\mathbb{Z}^n$ such that $\alpha \geq \beta$ implies $\alpha + \gamma \geq \beta + \gamma$.

- The standard example of a monomial order is the lexicographic ordering. They are specified via $x_1 > x_2 > \dots > x_n$, and multidegrees are compared by their first differing entry.

Definition 2. (Leading term, multidegree)

Fix a monomial ordering on $R = F[x_1, \dots, x_n]$.

- The leading term $LT(f)$ of $f \in R \setminus \{0\}$ is the monomial term of maximal order in f ; we set $LT(0) = 0$. The multidegree $\partial(f)$ is the multidegree of $LT(f)$.
 - $LT(fg) = LT(f)LT(g)$, $\partial(fg) = \partial(f) + \partial(g)$, and $\partial(f + g) \geq \max\{\partial(f), \partial(g)\}$ with equality if $\partial(f) \neq \partial(g)$.
- For an ideal $I \triangleleft R$, the ideal of leading terms is $LT(I) := (LT(f) \mid f \in I)$.

Definition 3. (Gröbner basis)

A Gröbner basis for a nonzero ideal $I \triangleleft R$ is a finite generating set $\{g_1, \dots, g_m\}$ for I whose leading terms generate $LT(I)$, i.e.

$$I = (g_1, \dots, g_m) \quad \text{and} \quad LT(I) = (LT(g_1), \dots, LT(g_m)).$$

Definition 4. (General Polynomial Division)

Fix a monomial ordering on R and let $g_1, \dots, g_m \in R \setminus \{0\}$. Given $f \in R$, initialize quotients $q_1, \dots, q_m$ and remainder r all to 0. Repeatedly examine the leading term $LT(f)$ of the current dividend:

1. If $LT(f)$ is divisible by some $LT(g_i)$, say $LT(f) = a_i LT(g_i)$ (taking the first such i), add a_i to q_i , replace f by $f - a_i g_i$ (which has lower order leading term), and reiterate.
2. If $LT(f)$ is divisible by no $LT(g_i)$, add $LT(f)$ to r , replace f by $f - LT(f)$, and reiterate.

The process terminates when the dividend is 0, yielding

$$f = q_1g_1 + \cdots + q_mg_m + r,$$

where each $q_i g_i$ has $\partial(q_i g_i) \leq \partial(f)$ and no nonzero term of r is divisible by any $LT(g_i)$.

Note that this is guaranteed to happen after finitely many steps due to the well-ordering of $\geq$.

We write $f \equiv r \pmod{G}$ for the remainder r produced by dividing f by $G = \{g_1, \dots, g_m\}$.

Theorem 5. (Division by a Gröbner basis)

Let $\{g_1, \dots, g_m\}$ be a Gröbner basis for the nonzero ideal $I \triangleleft R$. Then:

1. Every $f \in R$ may be written uniquely as $f = f_I + r$ with $f_I \in I$ and no nonzero monomial term of r divisible by any $LT(g_i)$.
2. Both f_I and r are computed by [Definition 4. \(General Polynomial Division\)](#), independent of the order in which the g_i are used.
3. r is the unique representative of the coset $f + I$ with the above property; in particular $f \in I \iff r = 0$.

Proof.

1. Existence follows from [Definition 4. \(General Polynomial Division\)](#).

For uniqueness, suppose that $f = f_I + r = f'_I + r'$. Then $r - r' = f'_I - f_I \in I$ so $LT(r - r') \in LT(I) = (LT(g_1), \dots, LT(g_m))$. But $LT(I)$ is a monomial ideal, so a polynomial is an element of $LT(I)$ iff each of its monomial terms are also in $LT(I)$. But none of the terms of r and r' is divisible by any $LT(g_i)$, so $r - r' = 0$.

2. Follows from (1).
3. Follows from (1). ■

Proposition 6. (Existence of Gröbner bases)

Let $I \triangleleft R$ be a nonzero ideal.

If $g_1, \dots, g_m \in I$ satisfy $LT(I) = (LT(g_1), \dots, LT(g_m))$, then $\{g_1, \dots, g_m\}$ is a Gröbner basis.

In particular, I always has a Gröbner basis.

Proof.

By Hilbert's Basis Theorem, take $g_1, \dots, g_m$ such that $LT(I) = (LT(g_1), \dots, LT(g_m))$.

Now if $f \in I$, then by [Theorem 5. \(Division by a Gröbner basis\)](#) we can write $f = \sum_i q_i g_i + r$ such that no term of r is divisible by $LT(g_i)$. But then $f \in I$ and $\sum q_i g_i \in I$ such that $r = 0$.

Hence the g_i 's generate I . ■

Definition 7. (S-polynomial)

For $f_1, f_2 \in R \setminus \{0\}$, let M be the monic least common multiple of $LT(f_1)$ and $LT(f_2)$. Define

$$S(f_1, f_2) := \frac{M}{LT(f_1)} f_1 - \frac{M}{LT(f_2)} f_2,$$

which is constructed such that the leading terms of f_1, f_2 cancel.

Lemma 8.

Suppose $f_1, \dots, f_m \in R$ all have the same multidegree α , and that the F -linear combination $h = a_1 f_1 + \dots + a_m f_m$ ($a_i \in F$) has strictly smaller multidegree. Then

$$h = \sum_{i=2}^m b_i S(f_{i-1}, f_i)$$

for some $b_i \in F$.

Proof.

Let f'_i denote the monic component of f_i such that $f_i = c_i f'_i$. Then

$$h = \sum a_i c_i f'_i = a_1 c_1 (f'_1 - f'_2) + (a_1 c_1 + a_2 c_2) (f'_2 - f'_3) + \dots + (a_1 c_1 + \dots + a_m c_m) f'_m.$$

Each $f'_{i-1} - f'_i = S(f_{i-1}, f_i)$ has multidegree strictly below α , as does h . Hence we have that $a_1 c_1 + \dots + a_m c_m = 0$ as it would otherwise introduce a degree α term. ■

Proposition 9. (Buchberger's Criterion)

Let $I = (g_1, \dots, g_m)$ be a nonzero ideal and set $G = \{g_1, \dots, g_m\}$. Then G is a Gröbner basis for I if and only if the remainder of $S(g_i, g_j)$ is 0 for all $1 \leq i < j \leq m$ on division by G .

Proof.

($\Rightarrow$) : Each $S(g_i, g_j) \in I$, so by [Theorem 5. \(Division by a Gröbner basis\)](#) its remainder on division by G is 0.

($\Leftarrow$) : Assume every $S(g_i, g_j) \equiv 0 \pmod{G}$. By [Proposition 6. \(Existence of Gröbner bases\)](#) it suffices to show $LT(I) = (LT(g_1), \dots, LT(g_m))$.

Take $f \in I$ and write $f = \sum_{i=1}^m h_i g_i$. Such a representation is not unique; so choose one that minimizes $\alpha := \max_i \partial(h_i g_i)$. Clearly $\partial(f) \leq \alpha$, and we also have

$$f = \sum_{\partial(h_i g_i) = \alpha} LT(h_i) g_i + \sum_{\partial(h_i g_i) < \alpha} (h_i - LT(h_i)) g_i + \sum_{\partial(h_i g_i) < \alpha} h_i g_i.$$

Suppose for a contradiction that $\partial(f) < \alpha$. The last two sums have multidegree $< \alpha$, so the first sum $\sum LT(h_i) g_i$ does too. Hence by [Lemma 8.](#) we can write

$$\sum_{\partial(h_i g_i) = \alpha} LT(h_i) g_i = \sum_{\partial(h_i g_i) = \alpha} a_i H_i g_i = \sum_{\partial(h_i g_i) = \alpha} b_i S(H'_{i-1} g_{i-1}, H'_i g_i)$$

where H'_i denotes the monic component of $LT(h_i)$ such that $LT(h_i) = a_i H'_i$. Note that H'_i is necessarily a monomial, such that $S(H'_{i-1} g_{i-1}, H'_i g_i)$ is equal to $S(g_{i-1}, g_i)$ multiplied by a monomial of multidegree $\alpha - \beta_{i-1,i}$; where $\beta_{i-1,i} := \partial(\text{lcm}(LT(g_{i-1}), LT(g_i)))$. But then $\partial(S(g_{i-1}, g_i)) < \beta_{i-1,i}$ as well; such that we can write $S(g_{i-1}, g_i) = \sum q_j g_j$ with $\partial(p_i g_i) < \beta_{i-1,i}$ via division by G . Hence, we see that $S(H'_{i-1} g_{i-1}, H'_i g_i) = \sum q'_j g_j$ for some q'_j with

$$\partial(p_i g_i) < \alpha.$$

But then this contradicts the minimality assumption on the multidegree α . So $\partial(f) = \alpha$, and

$$LT(f) = \sum_{\partial(h_i g_i) = \alpha} LT(h_i) LT(g_i) \in (LT(g_1), \dots, LT(g_m)).$$

Thus G is a Gröbner basis. ■

Definition 10. (Buchberger's Algorithm)

Let $I = (g_1, \dots, g_m)$ be a nonzero ideal. Then:

1. For each pair $i < j$, divide $S(g_i, g_j)$ by $G = \{g_1, \dots, g_m\}$ via [general polynomial division](#).
2. If every remainder is 0, then G is a Gröbner basis (by [Proposition 9. \(Buchberger's Criterion\)](#)).
3. If some $S(g_i, g_j)$ leaves a nonzero remainder r , append it to G and begin again.
Note that this process is guaranteed to terminate after finitely many steps due to the well-ordering of $\geq$.

Definition 11. (Minimal Gröbner basis)

A Gröbner basis $\{g_1, \dots, g_m\}$ is minimal if each $LT(g_i)$ is monic and $LT(g_j)$ is not divisible by $LT(g_i)$ for $i \neq j$.

By [Proposition 6. \(Existence of Gröbner bases\)](#), a minimal Gröbner basis can always be obtained from any Gröbner basis by discarding all g_j with $LT(g_j)$ divisible by another $LT(g_i)$.

- A minimal Gröbner basis is not unique, but its number of elements and its set of leading terms are: By [Proposition 6. \(Existence of Gröbner bases\)](#), G is a minimal Gröbner basis for I if and only if $LT(G)$ is a minimal generating set for $LT(I)$. But then minimal monomial generating sets for monomial ideals are unique.

Definition 12. (Reduced Gröbner basis)

A Gröbner basis $\{g_1, \dots, g_m\}$ for the nonzero ideal $I \triangleleft R$ is reduced if

1. each $LT(g_i)$ is monic; and
2. no term of g_j is divisible by $LT(g_i)$ for any $j \neq i$.

Every reduced Gröbner basis is minimal, and a reduced Gröbner basis can always be obtained from a minimal Gröbner basis by replacing each g_j with its remainder after division by the other elements; as the minimality guarantees that the remainder has the same leading term as g_j .

Theorem 13. (Uniqueness of reduced Gröbner basis)

Fix a monomial ordering on R . Then every nonzero ideal $I \triangleleft R$ has a unique reduced Gröbner basis.

Proof.

From [Definition 11. \(Minimal Gröbner basis\)](#), we have that any two reduced (hence minimal) bases have the same number of elements and the same leading terms, so write

$G = \{g_1, \dots, g_m\}$ and $G' = \{g'_1, \dots, g'_m\}$ with $LT(g_i) = LT(g'_i)$.

Fix i and put $f_i = g_i - g'_i \in I$. If $f_i \neq 0$ then $LT(f_i)$ is divisible by some $LT(g_j)$. For $j \neq i$, the reduced property forbids $LT(g_j)$ from dividing any term of g_i or g'_i , hence it cannot divide $LT(f_i)$. Moreover, $LT(f_i)$ also cannot be divisible by $LT(g_i)$ as all the terms of f_i has strictly smaller multidegree.

Hence $LT(f_i) = 0$, and $f_i = 0$. So $g_i = g'_i$ and $G = G'$. ■

Corollary 14.

Let $I, J \triangleleft R$. Then $I = J$ if and only if I and J have the same reduced Gröbner basis with respect to any one fixed monomial ordering.

Definition 15. (Elimination ideal)

For the ordering $x_1 > \dots > x_n$, the i -th elimination ideal of $I \triangleleft R$ is

$$I_i := I \cap F[x_{i+1}, \dots, x_n].$$

Proposition 16. (Elimination)

Let $G = \{g_1, \dots, g_m\}$ be a Gröbner basis for $0 \neq I \triangleleft R$ with respect to the lexicographic ordering $x_1 > \dots > x_n$. Then for each i , $G \cap F[x_{i+1}, \dots, x_n]$ is a Gröbner basis of the i th elimination ideal $I_i = I \cap F[x_{i+1}, \dots, x_n]$.

In particular, $I \cap F[x_{i+1}, \dots, x_n] = 0$ if and only if $G \cap F[x_{i+1}, \dots, x_n] = \emptyset$.

Proof.

Write $G_i = G \cap F[x_{i+1}, \dots, x_n] \subseteq I_i$. By [Proposition 6. \(Existence of Gröbner bases\)](#) it suffices to show $LT(G_i)$ generates $LT(I_i)$ as an ideal of $F[x_{i+1}, \dots, x_n]$.

$(\subseteq)$: That $(LT(G_i)) \subseteq LT(I_i)$ is clear.

$(\supseteq)$: Take $f \in I_i \subseteq I$. Since G is a Gröbner basis, write $LT(f) = \sum_j a_j LT(g_j)$ for $a_j \in R$. As $LT(f)$ is a monomial, we in fact see that $a_j \in F[x_{i+1}, \dots, x_n]$. Hence $LT(f)$ is a $F[x_{i+1}, \dots, x_n]$ -linear combination of the $LT(g_j)$'s that do not involve $x_1, \dots, x_i$. But then by the choice of the lexicographic ordering, if $LT(g_j)$ avoids $x_1, \dots, x_i$ then so does every term of g_j , i.e. $g_j \in G_i$.

Hence $LT(f) \in (LT(G_i))$. ■

Proposition 17. (Intersection of ideals)

Let $I, J \triangleleft R$. Then $tI + (1 - t)J$ is an ideal of $F[t, x_1, \dots, x_n]$ and

$$I \cap J = (tI + (1 - t)J) \cap F[x_1, \dots, x_n].$$

In particular, $I \cap J$ is the first elimination ideal of $tI + (1 - t)J$ with respect to

$$t > x_1 > \cdots > x_n.$$

This gives a concrete method of computing the intersection of two ideals.

Proof.

$(\subseteq)$: If $f \in I \cap J$, then $f = tf + (1 - t)f \in tI + (1 - t)J$.

$(\supseteq)$: Suppose $f = tf_1 + (1 - t)f_2 \in F[x_1, \dots, x_n]$ with $f_1 \in I$, $f_2 \in J$. Then $t(f_1 - f_2) = f - f_2 \in R$ such that $f_1 - f_2 = 0$, and $f = f_1 = f_2 \in I \cap J$. ■

Example 18. (n-colourings of graphs)

A finite graph $\mathcal{G}$ of size N is a set of vertices $1, \dots, N$ together with a collection of edges (i, j) . An n -coloring assigns one of n colors to each vertex so that adjacent vertices receive distinct colors.

Fix a field F with $|F| \geq n$ and a set $S \subseteq F$ of n distinct elements. Introduce a variable x_i per vertex. Then a coloring is an assignment $x_i = \alpha_i \in S$ with $\alpha_i \neq \alpha_j$ whenever (i, j) is an edge. Let

$$f(x) = \prod_{\alpha \in S} (x - \alpha) \in F[x], \quad g(x_i, x_j) = \frac{f(x_i) - f(x_j)}{x_i - x_j} \in F[x_i, x_j].$$

Then notice that $f(x_i) = 0$ enforces that $x_i \in S$, and $g(x_i, x_j) = 0$ enforces that $f(x_i) = f(x_j)$ and $x_i \neq x_j$.

- To see why $g(x_i, x_j) = 0$ implies $x_i \neq x_j$, it suffices to show that $g(x_i, x_i) \neq 0$. To that end, write $f = \sum_k c_k x^k$ and consider the identity $x_i^k - x_j^k = (x_i - x_j)(x_i^{k-1} + x_i^{k-2}x_j + \cdots + x_j^{k-1})$. Then

$$f(x_i) - f(x_j) = (x_i - x_j) g(x_i, x_j), \quad g(x_i, x_j) = \sum_k c_k (x_i^{k-1} + x_i^{k-2}x_j + \cdots + x_j^{k-1})$$

such that one sees that $g(x_i, x_i) = \sum k c_k x_i^{k-1} = f'(x_i)$; which is nonzero as f is separable by construction.

Hence n -colorings of $\mathcal{G}$ correspond bijectively to the solutions in F^N of the ideal $I \triangleleft F[x_1, \dots, x_N]$ given by the relations

$$\begin{cases} f(x_i) = 0 & 1 \leq i \leq N \\ g(x_i, x_j) = 0 & (i, j) \in \mathcal{G} \end{cases}.$$

By Hilbert's Nullstellensatz, the corresponding ideal I will have a solution in $\overline{F}^N$ if and only if I is a proper ideal. But then note that the $f(x_i) = 0$ condition enforces that solutions in $\overline{F}^N$ coincide exactly with solutions in F^N ; so the existence of n -colourings is equivalent to showing that the reduced Gröbner basis of I is not $\{1\}$. Furthermore, when an n -colouring does exist, solving the system via the method described in [Proposition 16. \(Elimination\)](#) explicitly gives all the possible n -colourings for $\mathcal{G}$.

Furthermore, we can choose F and S to optimize for computational savings:

- Take any F containing n distinct n th roots of unity such that we may take $f(x) = x^n - 1$ and $g(x_i, x_j) = \frac{x_i^n - x_j^n}{x_i - x_j}$.
- Take $S = F = \mathbb{F}_p$ in the special case when $n = p$. Then we have $f(x) = x^p - x$ and $g(x_i, x_j) = (x_i - x_j)^{p-1} - 1$.